WEEK -2

SORTING TECHNIQUES

1) Bubble Sort:

Program:

```
#include<stdio.h>
#define size 5
void bubblesort(int arr[], int n){
for(int i=0;i<n;i++){
 for(int j=0;j<n-i-1;j++){
  if(arr[j]>arr[j+1]){
   int temp=arr[j];
   arr[j]=arr[j+1];
   arr[j+1]=temp; }
 }
}
}
int main(){
 int arr[size];
 for(int i=0;i<size;i++){
  printf("enter element %d: ",i);
  scanf("%d",&arr[i]);
 }
 bubblesort(arr,size);
 for(int i=0;i<size;i++){
 printf("%d\n",arr[i]);  }
 return 0;
}
```

Output:

```
amma@amma11:~$ gcc bubble_sort.c -o bubble_sort
amma@amma11:~$ ./bubble_sort
enter element 0: 6
enter element 1: 9
enter element 2: 8
enter element 3: 1
enter element 4: 3
1
3
6
8
9
```

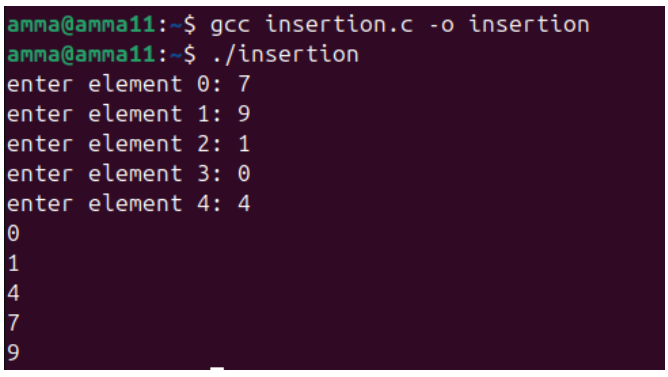
2) **Insertion Sort:**

Program:

```
#include<stdio.h>
#define size 5
void insertionsort(int arr[], int n){
for(int i=1;i<n;i++){
  int key=arr[i];
  int j=i-1;
  while(j>=0 && arr[j]>key){
   arr[j+1]=arr[j];
   j--;
  }
  arr[j+1]=key;
}
}

int main(){
 int arr[size];
 for(int i=0;i<size;i++){
  printf("enter element %d: ",i);
  scanf("%d",&arr[i]);
 }
 insertionsort(arr,size);
 for(int i=0;i<size;i++){
 printf("%d\n",arr[i]);
 }
 return 0;
}
```

Output:



3) **Selection Sort:**

Program:

```
#include<stdio.h>
#define size 5
void selectionsort(int arr[], int n){
for(int i=0;i<n-1;i++){
  int min=i;
  for(int j=i+1;j<n;j++){
   if(arr[j]<arr[min]){
    min=j;
   }
  }
  int temp=arr[i];
  arr[i]=arr[min];
  arr[min]=temp;
}
}
int main(){
 int arr[size];
 for(int i=0;i<size;i++){
  printf("enter element %d: ",i);
  scanf("%d",&arr[i]);
 }
 selectionsort(arr,size);
 for(int i=0;i<size;i++){
 printf("%d\n",arr[i]);
 }
 return 0;
}
```

Output:

```
amma@amma11:~$ gcc selection_sort.c -o selection_sort
amma@amma11:~$ ./selection_sort
enter element 0: 9
enter element 1: 1
enter element 2: 0
enter element 3: 3
enter element 4: 6
0
1
3
6
9
```

4) **Bucket Sort:**

Program:

```
#include<stdio.h>
#define size 5
void bucketsort(int arr[], int n){
 int bucket[10]={0};
 for(int i=0;i<n;i++){
  bucket[arr[i]]++;
 }
 int index=0;
 for(int i=0;i<10;i++){
  while(bucket[i]--){
   arr[index++]=i;
  }
 }
}
int main(){
 int arr[size];
 for(int i=0;i<size;i++){
  printf("enter element %d: ",i);
  scanf("%d",&arr[i]);
 }
 bucketsort(arr,size);
 for(int i=0;i<size;i++){
 printf("%d\n",arr[i]);
 }
 return 0;
}
```

Output:

```
amma@amma11:~$ gcc bucket_sort.c -o bucket_sort
amma@amma11:~$ ./bucket_sort
enter element 0: 8
enter element 1: 2
enter element 2: 0
enter element 3: 1
enter element 4: 7
0
1
2
7
8
```

5) **Heap Sort:**

Program:

```
#include<stdio.h>
#define size 5
void heapify(int arr[], int n, int i){
int largest=i;
int left=2*i+1;
int right=2*i+2;
if(left<n && arr[left] > arr[largest]){
 largest = left;
}
if(right<n && arr[right] > arr[largest]){
 largest = right;
}
if(largest!=i){
 int temp=arr[i];
 arr[i] = arr[largest];
 arr[largest]=temp;
 heapify(arr,n,largest);
}
}

void heapsort(int arr[], int n) {
   for (int i = n/2 - 1; i >= 0; i--) {
      heapify(arr, n, i);
   }
   for (int i = n - 1; i > 0; i--) {
      int temp = arr[0];
      arr[0] = arr[i];
      arr[i] = temp;
      heapify(arr, i, 0);
   }
}
int main() {
   int arr[size];
   for (int i = 0; i < size; i++) {
      printf("enter element %d: ", i);
```

```
    scanf("%d", &arr[i]);
  }
  heapsort(arr, size);
  for (int i = 0; i < size; i++) {
    printf("%d\n", arr[i]);
  }
  return 0;
}
```

Output: